nature microbiology

Review article

https://doi.org/10.1038/s41564-025-02103-7

The multi-kingdom cancer microbiome

Received: 6 May 2024

Accepted: 28 July 2025

Published online: 9 September 2025

Check for updates

Anders B. Dohlman [1], Xiangyu Pan [2,3], Laurence Zitvogel [4,5,6,7] & Iliyan D. Iliev [2,3,8,9]

Microbial influence on cancer development and therapeutic response is a growing area of cancer research. Although it is known that microorganisms can colonize certain tissues and contribute to tumour initiation, the use of deep sequencing technologies and computational pipelines has led to reports of multi-kingdom microbial communities in a growing list of cancer types. This has prompted discussions on the role and scope of microbial presence in cancer, while raising the possibility of microbiome-based diagnostic, prognostic and therapeutic tools. However, additional investigation and thorough validation of cancer microbiome findings are required before this translational potential can be realized. Here we provide historical context and a conceptual framework for the so-called cancer microbiome and summarize experimental studies into tumour-associated bacteria, fungi and other microorganisms. We also discuss the current evidence for microbial colonization of tumours and their varied influence on the disease, including recent debates. Finally, we consider outstanding questions and discuss our outlook for the field.

Our modern understanding of cancer is intertwined with the study of infectious disease (Fig. 1). In 1911, decades before we understood that human genomic changes were the underlying cause of cancer, Francis Peyton Rous demonstrated that an unidentified infectious agent could induce the development of sarcoma in chickens. He further showed that the persistence of what is now known as *Rous sarcoma virus* (RSV) at the tumour site[1] was essential for the cellular transformation of these tissues. At the time, the idea that cancer could be caused by a virus was largely dismissed, leading Peyton Rous to abandon this line of enquiry just a few years after his initial discovery.

In hindsight, the finding by Peyton Rous represented the first domino of many discoveries expanding our understanding of cancer, including the discovery of oncogenes. In 1976, well after nucleic acids were established as the basis of genetic heritability[2], Michael Bishop and Harold Varmus revisited Rous' findings, concluding that the RSV gene (*v-Src*) responsible for the transformation of host cells shared a homology with a gene conserved among their avian hosts (*c-Src*)[3], thereby solidifying the evidence that genetic alterations to endogenous cellular DNA were the underlying cause of cancer. The discovery of human oncogenes soon followed[3–5], beginning the era of cancer genomics. Meanwhile, other oncoviruses were discovered, including the Epstein–Barr virus (EBV), human papillomavirus (HPV) and human T cell lymphotropic virus (HTLV)[6–9].

The role of bacteria in cancer was revealed in 1983, when Barry Marshall and Robin Warren described an 'unidentified curved bacilli' in gastritis patients, a species today known as *Helicobacter pylori*[10]. It was assumed that bacteria could not survive in the highly acidic environment of the stomach, so Marshall famously swallowed the bacteria to prove he could culture it from a gastric endoscopy days later. In 2005, long after Marshall completed his antibiotic regimen, he and Warren were awarded the Nobel Prize for their work. Their findings represented the first concrete evidence that tumour-associated bacteria, as well as

[1]Department of Medical Oncology, Dana-Farber Cancer Institute, Boston, MA, USA. [2]Joan and Sanford I. Weill Department of Medicine, Gastroenterology and Hepatology Division, Weill Cornell Medicine, New York, NY, USA. [3]The Jill Roberts Institute for Research in Inflammatory Bowel Disease, Weill Cornell Medicine, New York, NY, USA. [4]Université Paris-Saclay, Faculté de Médecine, Le Kremlin-Bicêtre, France. [5]Gustave Roussy Cancer Campus (GRCC), Clinicobiome, Villejuif Cedex, France. [6]Institut National de la Santé Et de la Recherche Medicale (INSERM) U1015, Equipe Labellisée-Ligue Nationale contre le Cancer, Villejuif, France. [7]Center of Clinical Investigations in Biotherapies of Cancer (BIOTHERIS), Villejuif, France. [8]Department of Microbiology and Immunology, Weill Cornell Medicine, New York, NY, USA. [9]Immunology and Microbial Pathogenesis Program, Weill Cornell Graduate School of Medical Sciences, Weill Cornell Medicine, Cornell University, New York, NY, USA. e-mail: andersb_dohlman@dfci.harvard.edu; idi@ilievlab.org

https://doi.org/10.1038/s41564-025-02103-7

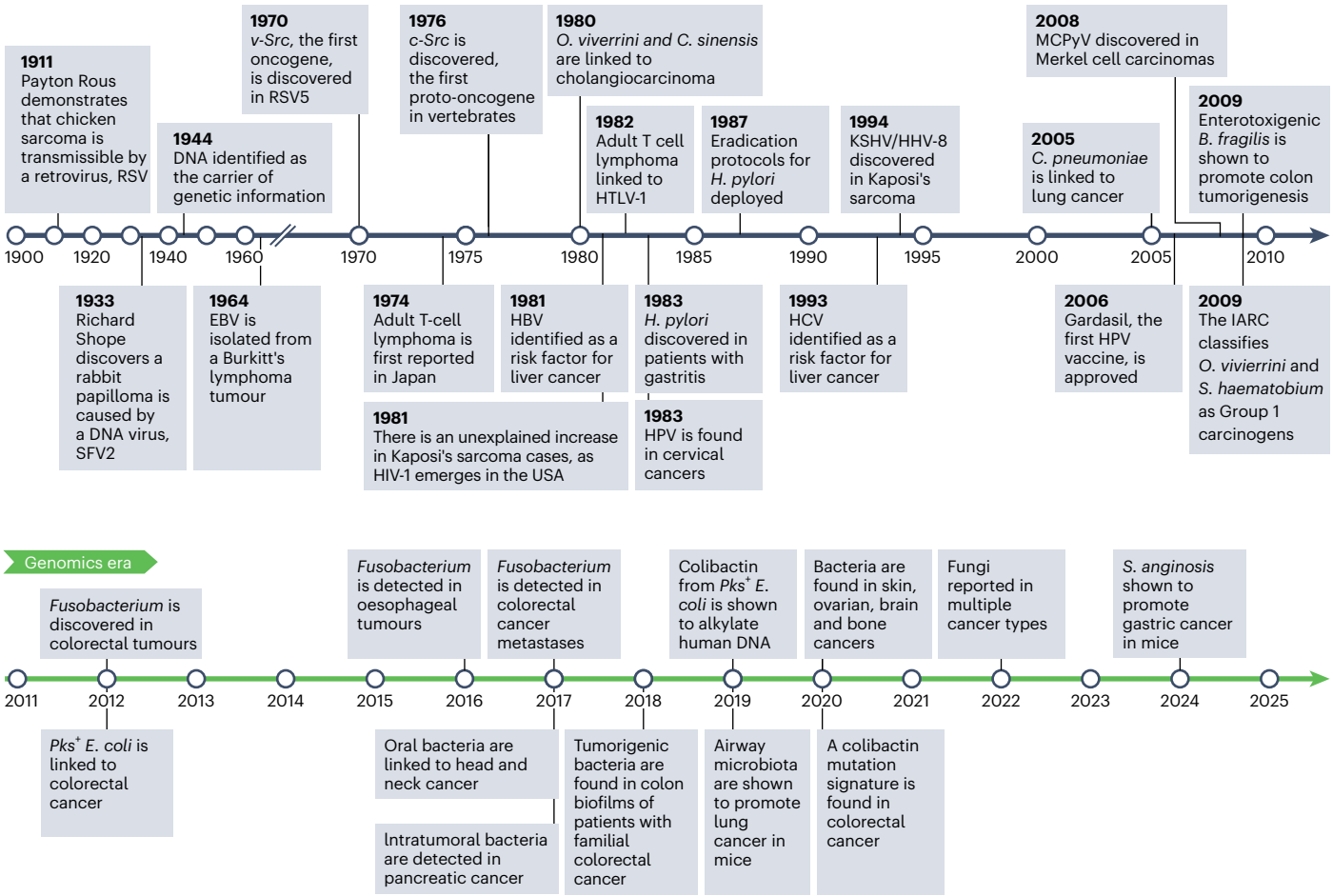

Fig. 1 | Timeline of discoveries related to cancer and microbiota. Broadly, cancer microbiome research can be organized into the pre-genomic era (1911–2009) and the post-genomic era (2010–present), reflecting the introduction of low-cost high-throughput sequencing technologies.

viruses, could also promote cancer. Today it is estimated that 89% of non-cardia gastric cancers (the most common form, affecting the lower part of the stomach) are attributable to *H. pylori* infection[11]. Meanwhile, public health efforts to reduce *H. pylori* infections have led to decreases in gastric cancer incidence by up to 50% in at-risk populations[12–14].

Over the past two decades, there has been a resurgence of interest in the role of microorganisms in cancer, particularly communities of bacteria and, more recently, fungi[15–22]. This is largely driven by the proliferation of low-cost genome sequencing, which has enabled the detection of tumour-associated microbial communities that previously evaded culture-based isolation techniques. Recent analyses of tumour sequencing data have suggested the presence of bacteria and fungi in association with many human cancer types, indicating that the presence of microbiota in cancer tissues could be more widespread than previously understood. However, there remains much to be learned about these microbiomes, including their colonization route and their influence on the tumour microenvironment.

Defining the cancer microbiome

The term 'cancer microbiome' and other similar terms are frequently used but not consistently defined. This is probably because microorganisms can influence cancer through a wide variety of mechanisms, and are known to act locally or systemically to exert harmful or beneficial effects on cancer, directly or indirectly (Fig. 2a)[23]. Following the precedent of the Human Microbiome Project[24], we define the 'cancer microbiome' as the collective genomes of microorganisms that are locally present at a tumour site, regardless of their influence on cancer development or progression.

Constituents of the broader human microbiome, in particular the gut microbiome, have been reported to influence tumorigenesis and therapeutic response[16,25–30], but as yet there is no conclusive evidence that the human commensal microbiome is directly involved in the aetiopathogenesis of cancer[31]. This Review will primarily focus instead on the cancer microbiome, as we define it, including the intracellular or extracellular microbes living among tumour cells or in the surrounding mucosa. Such tumour-associated species may be further subclassified by their degree of association with cancer and their localization around the tumour site (Fig. 2b and Box 1). For example, intratumoral microbes such as *Fusobacterium nucleatum* or infectious bacteria such as *Salmonella typhi* have been shown to persist intracellularly within tumour cells in colorectal[17,32] and gallbladder cancers[33], respectively. Meanwhile, mucosa-associated species such as *pks*$^+$ *Escherichia coli* and *Candida albicans* do not appear to live intratumorally but are instead found in the tumour-associated mucosa[21,26,34]. By contrast, although classified as Group 1 carcinogens[35], HTLV and HIV are not found in solid tumour tissues and instead promote cancer through systemic infection and immunosuppression[36,37]. Such pathogens are thus referred to as 'cancer-associated'. The localization of tumour-associated microorganisms at the tumour site is likely to have important implications for their role in tumour development and their influence on cancer therapy, as we will discuss. Other 'ambiguously defined' species have been reported in tumour tissues but have conflicting evidence regarding their presence at the cancer site, either due to being common contaminants (frequently identified in kit reagents or negative controls) or lacking sufficient orthogonal evidence or consistent replication in independent studies.

https://doi.org/10.1038/s41564-025-02103-7

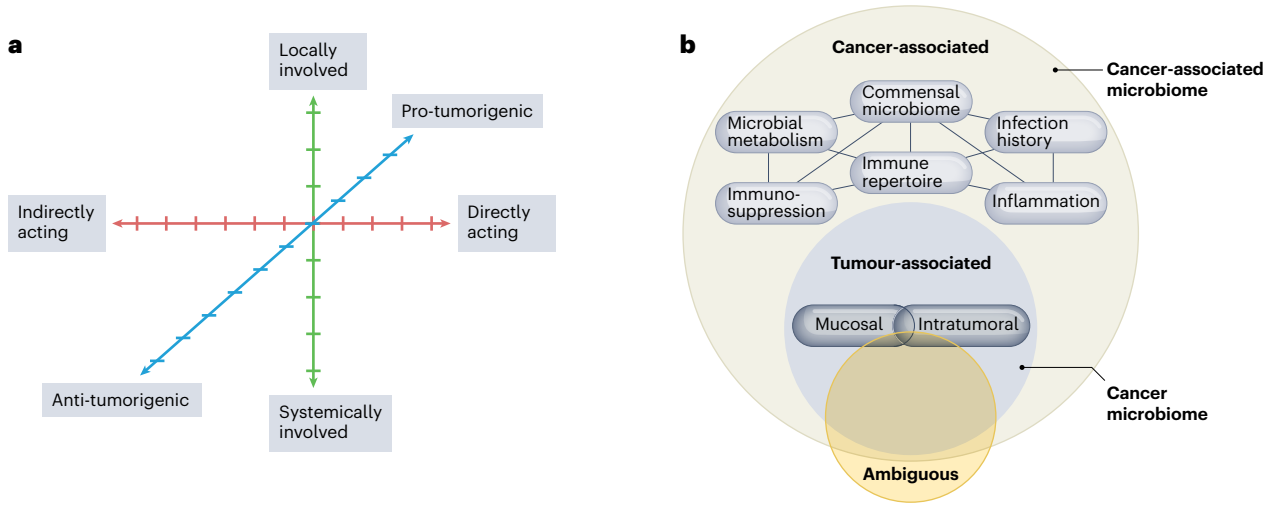

Fig. 2 | A conceptual framework for cancer–microbe associations. a, Cancer–microbe associations can broadly be classified along three axes. First, microbiota may promote oncogenesis directly (for example, by expressing oncoproteins or oncometabolites that transform the host cells they invade[27]) or indirectly (for example, by promoting a persistent inflammatory microenvironment through chronic infection)[23]. Second, cancer-associated microbiota may act locally, by promoting tumour development in the specific body site they colonize, or systemically, by promoting a general host phenotype that is susceptible to cancer. Third, microorganisms may exert a pro-tumorigenic, antitumorigenic or otherwise neutral effect on cancer. **b**, A conceptual framework relating definitions for the terms used to describe cancer–microbe associations (also see Box 1). 'Cancer-associated' describes all microbes that have been linked to cancer in some way, including microbes that influence cancer through systemic means (for example, by affecting metabolism or immunity), whereas 'tumour-associated' describes microbes that have been associated with the tumour site itself. Tumour-associated microbes can be subclassified as 'intratumoral' (embedded in the tumour itself), 'mucosal' (found in the tumour-associated mucosa) or 'ambiguous'.

Such species warrant careful interpretation before drawing conclusions regarding biological relevance.

A cancer microbiome is a collective of microbial genomes that may span multiple kingdoms. These communities each influence cancer through different mechanisms, but only the viruses HPV, EBV, Merkel cell polyomavirus (MCPyV) and Kaposi's sarcoma-associated herpesvirus (KSHV) are classified as directly oncogenic. However, numerous tumour-associated microbiota appear to promote cancer indirectly, spanning multiple kingdoms, including not just viruses, but also bacteria, fungi and parasitic flatworms. Although the mechanism is not well understood, interactions within these communities may influence cancer development. Both genomic and experimental studies have suggested intraspecies and intrakingdom functional interactions among cancer-associated microbes, including between bacteria and fungi. In oral tissues, *C. albicans* has been shown to interact antagonistically with *Lactobacillus* spp.[38,39] and cooperatively with *F. nucleatum*[40]. In gastric cancer, *H. pylori* and *Streptococcus anginosis* have been shown to act cooperatively to promote inflammation in mice[41,42]. In colorectal cancer, *F. nucleatum* has been shown to co-aggregate with a polymicrobial bacterial community including *Campylobacter*, *Gemella*, *Hungatella* and *Parvimonas* species, as well as others[43,44], while co-colonization by enterotoxigenic *Bacteroides fragilis* and *pks*$^+$ *E. coli* appears to have a pronounced effect on tumorigenesis compared to either bacteria alone[26]. Thus, further investigation is warranted into these interspecies and interkingdom relationships and their potential influence on cancer development.

Evidence for microbial colonization of tumours

A multi-kingdom spectrum of microorganisms is found in tumour sites throughout the human body (Table 1)[16,18,20–22,40,45–51]. Microbial colonization of tumour tissues can occur through a variety of mechanisms, although carriage can vary widely. HPV and EBV are ubiquitous in human populations, with infection occurring by sexual contact or oral transfer of saliva[52,53]. By contrast, KSHV and oncogenic parasites such as *Schistosoma haematobium* are endemic only to specific regions of the world[48,49]. Many cancer-associated bacteria and fungi are normally considered commensal members of the human microbiome, yet appear to promote cancer development under circumstances specific to the host, such as during chronic infection, immunosuppression or when certain strain-specific virulence factors are present[16,20,54]. An unresolved question for tumour-associated microorganisms is whether they are actively involved in cancer development or arrive after a suitable niche is established by tumorigenesis. Disentangling this 'passenger–driver' question[55] remains experimentally challenging in humans, as tumour-associated microorganisms are usually only identified after cancer has been diagnosed. Perhaps these two scenarios are not mutually exclusive—one intriguing possibility is that opportunistic colonization provides some growth advantage to the tumour, forming a symbiotic relationship between tumour-associated microbes and tumour cells[30].

In the following we delve into the specific evidence for microbial colonization of tumour tissues or tumour-associated areas by viruses, bacteria, fungi and parasites, each of which can promote cancer in distinct ways (Fig. 3). Additionally, we discuss specific mechanisms of colonization and evidence for their role in cancer development.

Viruses

Of the 11 'oncomicrobes' designated as Group 1 carcinogens by the International Agency for Research on Cancer (IARC), seven are viruses. Of these, four promote oncogenesis directly by expressing viral oncoproteins (HPV, MCPyV, EBV and HTLV-1) in precancerous tissues. For example, HPV expresses proteins that suppress p53 activity[56,57], and MCPyV expresses a T antigen that targets pRb[58]. HPVs comprise the most pervasive directly oncogenic viral family, responsible for around 90% and 70% of attributable risk for cervical and oropharyngeal cancer, respectively[52]. Among indirectly oncogenic, locally acting oncoviruses, hepatitis B virus (HBV) and hepatitis C virus (HCV) are the most prevalent. These viruses are associated with multiple cancer types but are most problematic in liver cancer, where 33% and 21% of mortality is attributable to inflammation from chronic HBC or HCV infections, respectively[59].

https://doi.org/10.1038/s41564-025-02103-7

BOX 1

Terms to describe microorganisms involved in cancer

Several terms are used to describe cancer–microbe associations, and these terms are frequently used inappropriately, inconsistently or interchangeably. Microorganisms that play a role in cancer or response to cancer therapies may be referred to as 'cancer-associated', 'tumour-associated', 'mucosa-associated' or 'intratumoral'. Here we provide definitions for these non-exclusive categorizations:

Cancer-associated
The microorganism has been linked to cancer in some way, but concrete evidence for its presence at the tumour site is lacking. Cancer-associated microbes may influence cancer or cancer therapy indirectly across body sites (for example, by affecting systemic immunity), but are not necessarily present at the tumour site.
Examples: intestinal *Bifidobacterium* influencing anti-PD-L1 efficacy in melanoma[139,183], commensal gut microbiota producing genotoxic metabolites[27] and *C. pneumoniae* in lung cancer[88].

Cancer-microbiome
A community of species that are locally present at the tumour site, including the tumour-associated mucosa.
Examples: the *Fusobacterium* co-abundance cluster in colorectal cancer[45] or the microbial association network in gastric cancer[42].

Tumour-associated
There is evidence that the organism co-localizes with tumours or sites near the tumour, such as the surrounding mucosa.
Examples: *S. anginosus* in gastric cancer[41,42] and *Blastomyces* in lung cancers[21].

Mucosa-associated
Several lines of evidence support the presence of the organism in the tumour-associated mucosa for cancer types with a mucosa layer.
Examples: *pks*$^+$ *E. coli* and enterotoxigenic *B. fragilis* in colorectal cancer[26,34], *H. pylori* in gastric cancer[70] and *C. albicans* in gastrointestinal cancers[21,22].

Intratumoral
Several lines of evidence indicate that the organism lives within the tumour itself. Some intratumoral microbiota enter tumour cells (intracellular), whereas others persist in the space between them (extracellular).
Example: *F. nucleatum* subsp. *animalis C2* in colorectal cancer[83,162,163].

Ambiguous
Species linked to tumour sites but with conflicting evidence regarding their presence in cancer, including mixed-evidence species that are both tumour-associated and recognized as common contaminants (for example, *E. coli*), as well as tumour-associated microbes that are pending validation in the absence of orthogonal evidence or consistent replication in independent studies (for example, *Acidovorax*, *Aspergillus* and *Malassezia*).

Systemic infection by oncoviruses can also increase cancer risk by targeting immune cells, as is the case for EBV in Burkitt's and Hodgkin's lymphomas and HTLV-1 in T cell lymphoma. Human immunodeficiency virus (HIV) promotes cancer indirectly, by causing systemic immunosuppression, thereby increasing the risk of infection by other oncoviruses such as KSHV. However, HTLV and HIV themselves are not found in solid tumour tissues.

Parasites

Although parasites are not always microbes in the strict sense, three of the 11 Group 1 infectious agents recognized IARC are endemic parasitic trematode flatworms[35]. Bile duct infections by *Opisthorchis viverrini* and *Clonorchis sinensis* together confer an 8–10% risk of cholangiocarcinoma[60], both of which are endemic in Southeast Asia and associated with the consumption of raw or undercooked fish. *Schistosoma haematobium*, also found in stagnant freshwater, is most common in Africa and the Middle East and primarily infects the bladders of infants and young children, often leading to bladder cancer later in life[61–63]. The carcinogenic mechanism of these parasites is not well understood, but it is believed to be indirect, via inflammation-induced fibrosis and DNA damage caused by the generation of oxidizing agents[47,48]. In endemic areas, these parasites are a pressing public health problem, as mortality for these cancers far exceeds the global average in these regions. Little is known about the extent to which these pathogens interact with other microorganisms at the tumour site.

Bacteria

Bacteria are by far the most abundant and diverse microbial kingdom in humans, with the vast majority residing in the oral cavity and gastrointestinal tract. Both mono- and polymicrobial communities of bacteria have been reported in various cancer types throughout the body, with varying degrees of evidence supporting their cancer-specific associations[21,22,45,64–68].

Currently, the only bacterium recognized as a Group 1 carcinogen by the IARC is *H. pylori*, for its role in stomach cancer[35]. *H. pylori* burrows through the gastric mucosa and into the gastric epithelium, where it promotes persistent inflammation[42,69,70]. If left untreated, the resulting inflammation can lead to chronic gastritis, mucosal atrophy, DNA damage and, ultimately, tumorigenesis. The tumorigenic effects of *H. pylori* are especially pronounced in strains expressing the *CagA* protein, the first bacterial toxin to be implicated in cancer. *CagA* promotes *H. pylori*'s oncogenic potential at least in part by enhancing its ability to persist in the highly acidic gastric microenvironment[71,72].

Unsurprisingly, the lower gastrointestinal tract, which harbours the highest concentration of bacteria in the human body, is linked to the most tumour-associated bacteria. Several species, including *pks*$^+$ *E. coli*, enterotoxigenic *B. fragilis*, *Campylobacter jejuni* and *Clostridium difficile* are known to express specific bacterial toxins that could promote colorectal cancer through a variety of proposed mechanisms, including Th17 signalling, DNA damage, cholesterol biosynthesis and the inhibition of the cytotoxic immune response[16,73–81]. Of these bacteria, *F. nucleatum* is typically the most abundant in colorectal tumours[32,45], and can persist in distal metastases[82]. Recent evidence suggests that a specific subclade of *F. nucleatum* dominates the colorectal cancer niche[83], suggesting potential adaptation. Many of these colorectal cancer-associated *F. nucleatum* strains express *fadA*[80], which appears to bind host E-cadherin to stimulate β-catenin signalling, and *fap2*, which has been shown to suppress the cytotoxic immune response[81,84].

Bacterial involvement has also been seen in other tumours of the digestive system. *Salmonella typhi* has substantial evidence linking it to gallbladder cancer risk, with higher rates of gallbladder cancer in

https://doi.org/10.1038/s41564-025-02103-7

Table 1 | Multi-kingdom tumour-associated microbes found at various body sites

	Viruses	Parasites	Bacteria	Fungi
Head and neck	HPV (oral)[a35,52] EBV (nasal)[a35]		*F. nucleatum*[187]	*Candida*[21,22]
Oesophageal			*F. nucleatum*[65,188]	*Cladosporium*[100] *Candida*[21,22,100]
Stomach	EBV[a35,53,189]		*H. pylori*[a10,35,41,42] *S. anginosis*[41,42]	*Candida*[21,22]
Large intestine			*F. nucleatum*[162,163] Enterotoxigenic *B. fragilis*[15,26,74,190] *pks*$^+$ *E. coli*[25,26,34,75,191] *Parvimonas micra*[192,193] *C. jejuni*[77] *Peptostreptococcus anaerobius*[78,79,194]	*Candida*[21,22]
Skin	KSHV[a35,195] MCPyV[35,195]			
Brain	JCV[b196–198]			
Lung			*C. pneumoniae*[88]	*Aspergillus*[199] *Blastomyces*[21,22]
Liver	HBV[a]/HCV[a35,200]			
Bile duct		*O. viverrini*[a35,47,48] *C. sinensis*[a35,47,48]		
Gallbladder			*S. typhi*[85]	
Pancreas				*Malassezia*[b201]
Bladder		*S. haematobium*[a35,47,48]		
Cervical	HPV[a9,35,52]			

Viruses, parasites, bacteria and fungi with evidence supporting their presence in tumour tissues and involvement in promoting cancer development. [a]Indicates an organism designated as Group 1 carcinogens by the IARC. [b]Indicates an organism whose role in cancer remains disputed[107,108].

areas endemic for typhoid *Salmonella*[85]. In pancreatic cancer, there is mixed evidence for the role of tumour-associated bacteria. Some studies have identified bacteria predictive of therapeutic resistance[86] and survival[86] in pancreatic cancer, whereas others have found that the pancreatic tumour microbiome is low-biomass and does not differ from normal adjacent tissues[87].

Cancers at other barrier surfaces have also shown evidence of microbial colonization, including malignancies of the lung, breast and skin. In lungs, there is epidemiological evidence that local *Chlamydia pneumoniae* infections confer risk for non-small cell lung cancer[88], and commensal airway bacteria have been shown to promote lung cancer development in mice by activating γδ T cells[89] or Th17 cells[90,91]. Tumour-associated bacteria have also been linked to lymphocyte infiltration in cutaneous melanoma[92], and in breast cancers there is evidence in mice that intratumoral bacteria can promote metastases to the lung[93].

Targeting tumour-associated bacteria may be a strategy for enhancing therapeutic outcomes for patients with cancer. A retrospective analysis by Wang and colleagues[94] found that the neoadjuvant use of antibiotics approved for treating *F. nucleatum* reduced the risk of recurrence or death by 25.5% in patients with colorectal cancer. They also showed that local liposome-based delivery of tinidazole eliminated tumour-associated *F. nucleatum* and conferred antitumour effects in mice. Targeting *F. nucleatum* in this way resulted in the release of calreticulin, an endoplasmic reticulum stress response molecular pattern of immunogenic tumour cell death, culminating in T cell priming and memory CD8$^+$ T cell responses against microbial and tumoral antigens. Hence, liposomal antibiotics may be effective for eradicating infected and non-infected tumour lesions. Somewhat paradoxically, the presence of intratumoral *F. nucleatum* has also been linked to improved response to immune checkpoint blockade in patients[95,96]. However, *F. nucleatum* lost this effect after knockout of a gene involved in producing butyrate[95], suggesting that it may have context- or strain-specific effects on the tumours it colonizes.

Fungi

Relative to bacteria, fungi are a small but highly immunoreactive component of the human gut and skin microbiomes, which are largely dominated by *Candida*, *Saccharomyces* and *Malassezia*. Epidemiological studies have linked fungi to cancer, especially in oesophageal and gastric malignancies[97,98], where recurrent fungal infections may contribute to carcinogenesis indirectly via chronic inflammation[50]. Studies have consistently found *Candida* species, predominantly *C. albicans*, in biopsies from patients with mild to severe oesophageal dysplasia[99] and a higher frequency of candidiasis in patients with oesophageal carcinoma than in those with oesophagitis[97]. More recent studies have explored the causative link between oesophageal cancer development and fungal colonization in animal models. Oral infection with *Cladosporium cladosporioides* was shown to promote oesophageal tumorigenesis in IKKα kinase-deficient mice with autoreactive T cells[100], while *Cladosporium* spp. and *C. albicans* have both been detected in association with biopsies from patients with oesophageal squamous cell carcinoma[100].

Candida has also been reported in colorectal adenomas[101]. Many *Candida* species are believed to produce acetaldehydes, which are genotoxic[102]. *Candida* also produce the toxin candidalysin, which is known to disrupt epithelial junctions[103] and directly damage intestinal cells in patients with ulcerative colitis[104], a risk factor for colorectal cancer. The SYK-CARD9 signalling axis has been implicated in fungi-mediated inflammasome activation and protection against colorectal cancer[105], whereas in CARD9-deficient mouse models, *Candida tropicalis* has been shown to promote colorectal cancer development through the induction of myeloid-derived suppressor cells[101]. However, the extent to which fungi affect the processes of colorectal cancer development in humans remains to be determined.

Additionally, fungi have been linked to pancreatic cancer. In mouse models, *C. albicans* has been shown to accelerate pancreatic tumorigenesis, and antifungal treatment increased responsiveness to

https://doi.org/10.1038/s41564-025-02103-7

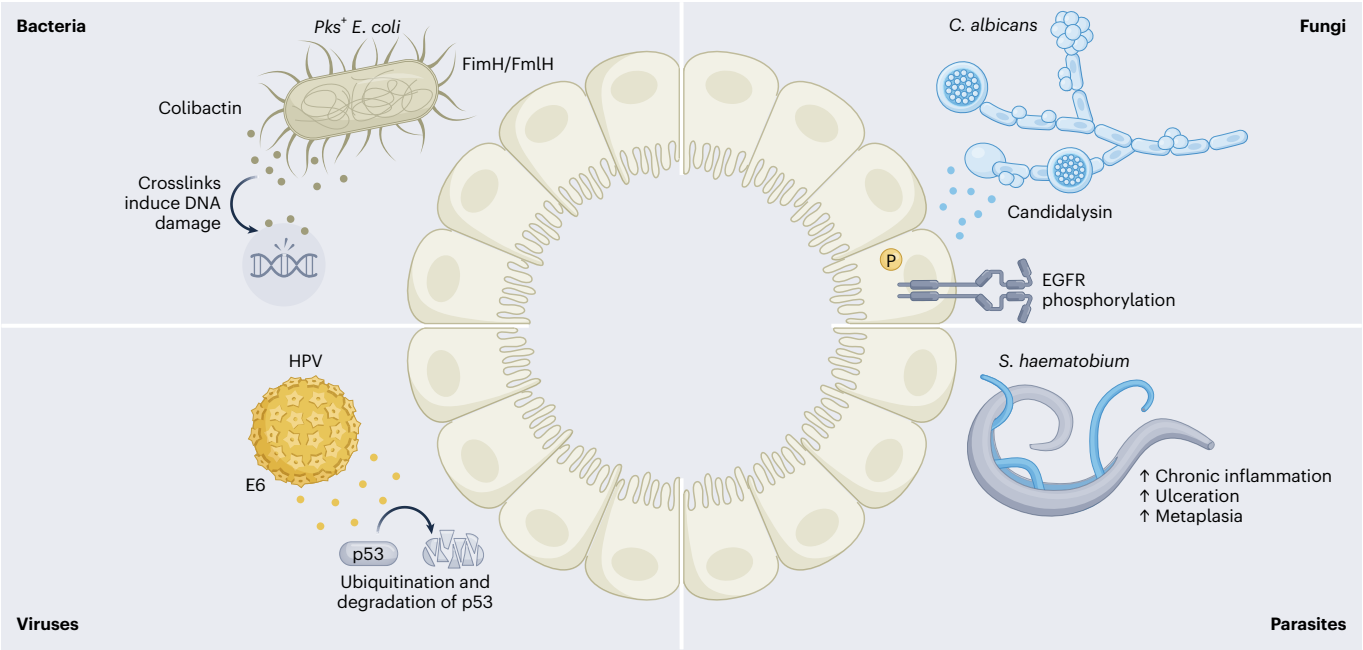

Fig. 3 | Tumour-associated microbes across kingdoms influence tumorigenesis through a variety of mechanisms. Representative species from each kingdom are shown. The bacterium *pks*$^+$ *E. coli* has been shown to secrete colibactin into host cells after binding them with FimH/FmlH[73], forming DNA crosslinks and DNA double strand breaks, which create a unique mutational signature[25,29,191]. The fungus *C. albicans* has been shown to promote epidermal growth factor receptor (EGFR) phosphorylation and the innate epithelial immune response[103,104,202,203]. Viral strains of HPV integrate into the genomes of their host cells and express the protein E6, which ubiquitinates p53 for degradation by the proteasome[56,57]. The parasite *S. haematobium* embeds itself into the bladder of its host, promoting chronic inflammation, ulceration and metaplasia, leading to bladder cancer[61–63].

radiotherapy. Interestingly, Dectin-1, a C-type lectin receptor involved in fungal cell wall β-glucan recognition, has been shown to accelerate tumorigenesis in mouse models of pancreatic cancer[106]. In a mouse model of pancreatic ductal adenocarcinoma, oncogenic Kras G12D mutations increased interleukin-33 (IL-33) expression, leading to activation of intratumoral type 2 innate lymphoid cells, which was further augmented by repeated gavage with *Alternaria alternata* and *Malassezia globosa*[28]. *Malassezia*, a skin commensal, is reported to be enriched in pancreatic tumours compared to normal pancreatic tissues in humans and mouse models, with colonization with a strain of *M. globosa* augmenting the incidence of pancreatic tumours[17]. However, the role of *Malassezia* in pancreatic cancer remains contested[107,108], and it has yet to be conclusively established whether *Malassezia* contributes to pancreatic cancer development and whether/how it arrives at the tumour site.

Tumour-associated microbes and metastasis

After colonizing the tumour site, microbiota may affect metastasis[109] by influencing the epithelial-to-mesenchymal transition (EMT), the expression of cancer stem cell (CSC) markers, the tumour microenvironment and the formation of extracellular vesicles (EVs), each of which can enable the metastasis of cancer cells to distant sites[110–112] (Fig. 4).

Several studies have linked microorganisms to EMT[113]. During EMT, normal epithelial cells take on the properties of mesenchymal cells, including increased motility, reduced cellular adhesion, and resistance to apoptosis, leading to increased metastatic potential. *F. nucleatum* produces a protein called *Fn-Dps* that has been shown to enhance EMT in mouse models of colorectal cancer by upregulating chemokines such as CCL2 and CCL7, leading to metastasis[113]. Another study found that certain bacteria, including *E. coli*, *Oscillatoria* species and the butyrate-producing bacteria SM4/1 are associated with increased expression of EMT-related genes in bladder cancer[114]. Microorganisms can also upregulate EMT-associated transcription factors, as seen

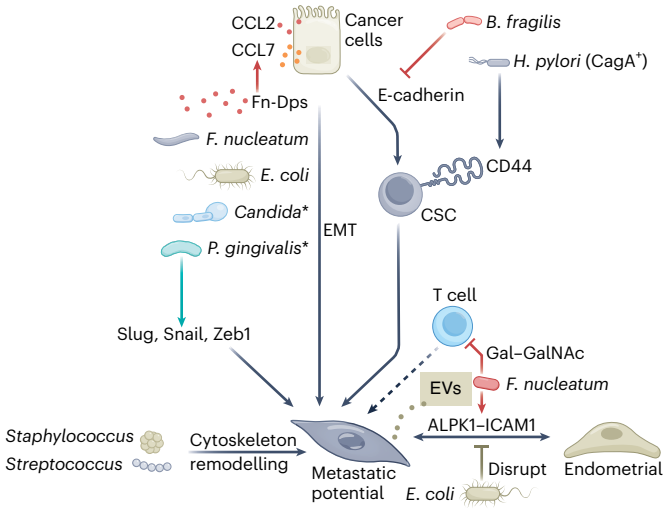

Fig. 4 | Potential pathways by which microbiota might influence cancer metastasis. Microorganisms may influence metastasis by promoting EMT, the expression of CSC markers, cytoskeletal remodelling or through the production of EVs. Asterisks indicate associations that have not yet been confirmed with direct experimental evidence.

with *Porphyromonas gingivalis*, which increases levels of Slug, Snail and Zeb1, enhancing the migration of oral epithelial cells in vitro[115]. Meanwhile, in human colon tumour mucosa, the presence of *Candida* has been linked to the downregulation of adhesion-related genes, suggesting a potential contribution to metastasis[21].

In addition to affecting EMT, microbiota can also influence the expression of CSC markers, which are associated with enhanced metastatic potential. In gastric cancer, *H. pylori* infection, particularly by

CagA-positive strains, triggers EMT-like changes while increasing the expression of CD44, a CSC marker, leading to greater tumour invasion and sphere formation[116]. In mouse models of breast cancer, the *B. fragilis* toxin has been shown to cleave E-cadherin, leading to β-catenin nuclear localization and increased stemness, tumour growth and metastatic progression[117].

Preclinical evidence suggests that microbiota may also drive the tumour microenvironment towards a state that supports cancer metastasis, particularly in the later stages of disease. In mouse models of colorectal cancer, *F. nucleatum* has been shown to activate the NF-κB signalling pathway via the ALPK1 receptor, promoting endothelial cell adhesion and metastasis[118], while tumour-associated *E. coli* has been linked to disruption of the gut vascular barrier, facilitating bacterial and metastatic spread to the liver in mice[119]. Similarly, *C. jejuni* has been shown to promote colorectal cancer metastasis in mice through the expression of a cytolethan distending toxin *cdtB*, promoting JAK-STAT signalling[120]. In breast cancer, *F. nucleatum* suppressed immune responses in mice by reducing tumour-infiltrating T cells through the abundant Gal-GalNAc, aiding metastasis[121]. In a mouse model of spontaneous breast tumours, *Staphylococcus*, *Lactobacillus* and *Streptococcus* were highly abundant and shown to inhibit the RhoA-ROCK signalling pathway, leading to actin cytoskeleton remodelling, thereby promoting metastasis[93].

Finally, recent work has shown that tumour-associated microbiota can promote metastasis through the formation of EVs, which can modulate pro-inflammatory signalling, immunosuppression and communication between cancer cells and their environment[122,123]. In breast cancer models, *F. nucleatum*-derived EVs have been shown to enhance tumour migration and invasion via TLR4[124].

Thus far, most studies have largely focused on bacterial contributions to the metastatic process, which can occur through various mechanisms and at various disease stages. Much less is currently known about fungi in relation to metastatic processes. Studies on potential fungal interactions with other microorganisms might shed light on how the multi-kingdom cancer microbiome influences metastasis.

Antitumorigenic influence of the cancer microbiome

There is evidence that certain microorganisms may be capable of preventing cancer development and progression. Most research in this area has focused on long-range interactions between gut bacteria and cancer development at distant body sites. For example, numerous studies have linked members of the human gut microbiome to improved response rates to immune checkpoint blockade[125] in the context of lung cancer[126–129], melanoma[130–132], renal cancer[133,134], bladder cancer[126,134] and colorectal cancer[135], raising the possibility that the human microbiome could be modulated to enhance a therapeutic response through faecal microbiota transplantation (FMT)[136], probiotics[137], antibiotics[16,18] and antifungal therapeutics[138]. In particular, FMT has shown promise in both preclinical[132,139] and clinical[140] studies of melanoma. These results suggest that the human microbiome may mitigate cancer development and progression systemically, suggesting opportunities for microbiome-based therapeutic interventions.

However, tumour-associated microbiota may also exert a beneficial effect at the local level, by mitigating tumour development, by enhancing the therapeutic response or by promoting the antitumour immune response. For example, tumour-associated microbial diversity has been linked to improved responses to immunotherapy in pancreatic cancer[68], whereas in colon cancer, mucosa-associated *Holdemania biformis* has been shown to reduce tumorigenesis in mice through the production of short-chain fatty acids[141]. However, despite these promising findings, identifying cancer-preventive tissue-associated microorganisms remains a challenge due to a lack of longitudinal studies that sample pre-tumorigenic tissue sites. Additional research is required to better understand how the endogenous microbial composition of human tissues could reduce the risk of cancer development at those sites.

Nevertheless, there is evidence that the delivery of exogenous anti-tumorigenic microorganisms to the tumour site after diagnosis could be an effective therapeutic strategy. Bacterial immunotherapy as a treatment for cancer has been attempted throughout history, dating from early antiquity to William Coley's controversial experiments in the late nineteenth century[142,143]. Presently, three oncolytic microbial therapies are approved by the US Food and Drug Administration (FDA). Perhaps the most successful of these is the attenuated *Mycobacterium bovis*-based Bacillus Calmette–Guérin (BCG) vaccine, whose local administration to bladder tumours has been standard-of-care for 35 years, with a 90% success rate[144,145]. An oncolytic viral therapy based on *Talimogene laherparepvec* (TVEC), a type-1 HSV genetically modified to enhance major histocompatibility complex (MHC) loading and tumour antigen presentation by dendritic cells[146], was approved for advanced melanoma in 2015, while the third is Nadofaragene firadenovec-vncg, an engineered adenovirus for BCG-resistant bladder cancer[147].

Bacteria engineered to target the tumour site and provoke an antitumour immune response or deliver therapeutic payloads have also gained interest[148–150]. In mice, attenuated strains of *Salmonella typhimurium* have been shown to selectively localize to and expand within tumours[151,152], making them an attractive vehicle for intracellular or extracellular delivery of therapeutics agents[153–157]. Similar engineering approaches have also been applied to *E. coli*, including cyclic intratumoral release of chemotherapeutics[158], cytokine-expressing payloads[159] and therapy-enhancing metabolites[160]. Most recently, *S. epidermidis*, a common skin commensal, was engineered to express tumour antigens in mice, eliciting an antitumour immune response in local and metastatic lesions[161].

Such reports suggest that both endogenous and exogenous microorganisms could be harnessed to mitigate cancer development and progression. Although bacteria and viruses have been examined extensively for their potential anti-tumorigenic effects, fungi remain underutilized, despite growing evidence linking them to cancer. Fungi are highly immunogenic, with an expansive genomic toolbox[19], suggesting potential anti-tumorigenic approaches akin to those deployed for bacteria and viruses. As our knowledge of tumour microbiome biodiversity continues to grow across kingdoms, so does the scope of potential microbiome-based therapeutic interventions.

Spatial heterogeneity in cancer microbiomes

Despite growing evidence for cancer-specific microbial signals[21,22,45,67], substantial gaps remain in our understanding of the biodiversity and biogeographical distribution of microorganisms within and across cancer types, as well as the routes by which they arrive at the tumour site. Thus, future work will involve charting the spatial distribution of microorganisms within and across body sites as well as investigating the extent to which tumour-associated microorganisms shape, or are shaped by, intratumoral heterogeneity.

Across cancer types, there is a clear variation in microbial colonization (Table 1). However, microorganisms also exhibit variable colonization patterns within the cancer site itself. For example, in colorectal cancer, *F. nucleatum* is enriched at the tumour site compared to the surrounding tissue[162,163], which may be explained by selective pressure within the colorectal cancer niche[83]. Meanwhile, in stomach cancer, *H. pylori* is frequently found at lower abundance at the tumour than in the surrounding tissue – as *H. pylori*-infected gastric lesions progress towards malignancy, the tumour microenvironment seems to become less favourable for *H. pylori* growth[69]. Determining the changing colonization patterns of tumour-associated microorganisms during cancer progression will be important for understanding and disrupting their role in the disease.

Although *F. nucleatum* and *H. pylori* are known to exist within the tumour epithelium, some cancer-associated microorganisms of the lower gastrointestinal tract instead aggregate in biofilms at the mucosal surface of the intestinal lumen. Despite not coming into direct contact with the tumour, these microbes can influence the tumour-immune microenvironment from the surrounding mucosa. For example, *pks*$^+$ *E. coli* has been linked to mutational signatures in colorectal cancer[25], but is associated with the intestinal mucosa rather than the tumour itself[34]. Within the colonic mucosa, *E. coli* and *B. fragilis* have been shown to behave cooperatively in biofilms, secreting oncotoxins that promote inflammation and DNA damage[26]. Similarly, *C. albicans* can influence intestinal inflammatory processes in a strain-dependent manner[104], but is associated with the mucosa surrounding the tumour and is not intratumoral[21]. However, it is not yet known whether other tumour-associated species influence cancer from within the tumour or from the surrounding mucosa.

Across tumour sites, cancer-type specific colonization is presumably determined by tissue-specific baseline exposure to exogenous microorganisms, variable immune surveillance of different tissue types, the molecular composition of the tumour microenvironment, and the metabolic requirements of the cancer itself. Understanding cancer-site specificity of microbial colonization will require applying multi-omic approaches[164] to simultaneously measure microbial composition, metabolic function and immune infiltration and will be important for understanding organ-specific tumour–microbe interactions.

Within cancer sites, the spatial distribution of tumour-associated microorganisms cannot be evaluated using bulk sequencing data. However, single-cell and spatial transcriptomic and proteomic approaches show promise in clarifying the biogeographical localization of microbes within a heterogeneous tumour-immune microenvironment. Galeano-Niño and colleagues[64] deployed these technologies in colon and oral tumours to demonstrate that bacterial communities aggregate in immunosuppressed microniches with low vascularization, suggesting non-random microbial colonization of the tumour site. Although the mechanisms of these patterns remain incompletely understood, it is apparent that as spatial techniques mature, they will continue to provide insight into the distribution of microorganisms within the tumour site.

Pan-cancer analyses suggest bacterial and fungal communities are present in many cancer types

Recently, studies have sought to characterize tumour-associated microbial communities in multiple cancer types simultaneously. In the first of these studies, Nejman and colleagues[67] reported cancer-type-specific intratumoral bacteria in seven cancer types (breast, lung, ovary, pancreas, melanoma, bone and brain) using targeted sequencing and imaging, challenging conventional knowledge about the scope of microbial colonization across body sites.

In a pan-cancer analysis of tumour metastases, Battaglia and colleagues[46] found that the bacterial composition of metastatic tumours is determined to a greater degree by the location of the metastasis rather than the primary tumour site, while also being influenced by microenvironmental features such as hypoxia and immune infiltration[46]. Given that other preclinical studies have shown that some intratumoral bacteria can persist in distal metastases[82,93], future studies will be needed to investigate the mechanisms by which microorganisms colonize metastatic and pre-metastatic sites and the ways in which the microbial composition and microenvironment interact to influence these processes. Moreover, it remains unclear whether the failure of the local or systemic immune system is a cause or a consequence of bacterial presence in tumours.

Other pan-cancer analyses have sought to characterize tumour-associated fungi. In 2022, two independent articles reported fungal DNA in several cancer types. Dohlman and colleagues[21] identified the involvement of *Candida* and *Saccharomyces* spp. in three gastrointestinal cancer types and *Blastomyces* in a subset of lung cancers[21], whereas Narunsky-Haziza and colleagues[22] reported a more diverse fungal presence in 35 cancer types using The Cancer Genome Atlas (TCGA) and other datasets. These studies differed in design, scope, decontamination and validation approaches, focusing on different cancer types and regions within the tumour environment (mucosa-associated[21] versus intratumoral[22]), which may collectively explain the discrepant results. However, both demonstrated that fungi and bacteria are frequently found together, suggesting the possibility of cross-kingdom interactions that may affect the tumour microenvironment.

Some of these pan-cancer studies have focused on tumour tissues in body sites that were not previously believed to harbour microbiota. How bacteria colonize putatively sterile sites remains unknown. The transient entry of microbiota directly into the bloodstream has been posited as one possible mechanism for colonization of some cancers[165]. It has also been proposed that microorganisms could translocate into the bloodstream as Trojan horses within dendritic cells[166] due to impaired epithelial barrier function resulting from cancer development or overt immunostimulation. However, there is not yet conclusive evidence for either of these models, and the extent to which microbiota are able to persist in the bloodstream of healthy individuals remains disputed[167].

Challenges of studying tumour-associated microorganisms with tumour sequencing data

Tumour sequencing databases are a powerful resource for studying tumour-associated microorganisms. When tumour samples are sequenced for genomic profiling, genetic material from tumour-associated microorganisms is also present. Owing to large sample numbers and longitudinal clinical annotations, these collections can provide valuable insights into tumour-associated microorganisms in human cancers. However, because these studies are not usually designed to recover microbial sequences, their use for microbiome analysis requires special considerations. Studies using such datasets have reported microorganisms at traditionally sterile tumour sites, prompting debates about the handling of batch effects and the use of microbial draft genomes[107,108,168–172].

This discourse has revealed that detection of bona fide microbial signals in large-scale sequencing datasets presents considerable challenges, and unstandardized handling of large-scale sequencing datasets can lead to spurious signals of various types. In particular, it has become clear that biological false positives, arising from laboratory contamination, and technical false positives, arising from the misclassification of non-microbial sequences, can produce erroneous microbial signatures and conclusions[45,173–177]. The recognition of these potential pitfalls highlights the need for a standardized framework for metagenomic analyses of bulk tumour sequencing datasets. In this section and in Box 2, we propose guidelines for mitigating biological and technical contamination in microbiome analyses of these datasets.

Controlling biological contamination is of particular importance for analyses of tumour sequencing datasets, as the microbial biomass of these tissues is generally low compared to other common metagenomic analytes such as stool. Low-biomass tissues are especially susceptible to biological contamination, necessitating rigorous study design and transparent reporting practices, as emphasized in a recent consensus statement by leading microbiome researchers[178]. Among their recommendations is the inclusion of negative controls at multiple steps during sample preparation and sequencing; however, such controls are rarely incorporated into tumour sequencing pipelines. Because most computational tools for decontamination rely on such controls, it is necessary to deploy de novo techniques for identifying and removing contamination in these studies. Blood samples or other tissues, which are not expected to harbour a significant microbial presence and are

https://doi.org/10.1038/s41564-025-02103-7

BOX 2

Guidelines for mitigating false-positive microbial signals in tumour sequencing datasets

Microbiome-based studies of tumour sequencing data involve searching for microbial DNA or RNA in sequenced tumour tissues. Although this can be a powerful approach for studying tumour-associated microbes, these analyses can produce spurious signals due to biological and technical false positives. Here we provide guidelines for mitigating these issues.

Strategies for minimizing technical false positives

When searching for microorganisms in tumour sequencing data, technical false positives occur when non-microbial sequences are mistakenly aligned to microbial genomes.

1. Preliminary subtraction of human sequences: before searching for microbial reads, steps should be taken to remove host-aligned sequences, accounting for incomplete human genome assemblies and variable regions of the human genome and/or transcriptome.
2. Quality control of putatively non-human sequences: technical false positives can be reduced by filtering out reads with excessive low-quality or low-complexity sequences. Masking and/or trimming of adaptor sequences and low-complexity sequences is advised.
3. Vetting microbial reference genomes: the microbial reference genomes used for alignment should be carefully vetted for potential human contamination. Complete genomes should be used when possible, and draft genomes should be avoided.
4. Inclusion of a host genome during alignment: even after host sequences are removed, including a copy of the human genome alongside microbial reference genomes allows each read the opportunity to align to the correct genome, reducing misalignment errors.
5. Validate microbial genome alignments: false-positive alignments usually do not occur at random across microbial genomes but tend to cluster in specific regions[204]. Microbial genome alignments should be spot-checked when there is suspicion of false positives.
6. Use of more than one classification tool: several tools are available for metagenomic classification (for example, Kraken2, PathSeq and MetaPhlAn), each with different methodological implementations that come with corresponding strengths and weaknesses. Cross-validating results with multiple tools can improve reproducibility across methods.
7. Prior knowledge and biological plausibility: detected microbial species should be evaluated in the context of the available literature, especially their isolation source and growth requirements. Microbiota that have not previously been detected in the tissue type of interest require additional scrutiny.

Strategies for mitigating biological false positives

Biological contaminants can be introduced from the laboratory environment or reagents during sample preparation.

1. Sterile technique: if you are doing the DNA extraction and library preparation yourself, take care to avoid contaminating the sample at every step. Perform experiments in a sterilized fume hood.
2. Negative controls: wherever possible, include negative controls so contamination can be identified in sequencing data later on[174]. These can include DNA extraction controls as well as non-template controls[178].
3. Inclusion of normal tissue controls: simultaneous collection of blood samples, sterile tissues and adjacent normal tissues alongside tumour samples provide biological controls for identifying contamination introduced during biopsy. During analysis, such samples can also be helpful for identifying tumour- or normal-tissue enriched taxa.
4. Study batch effects: for retrospective analyses of tumour sequencing data, negative controls may not be available. Performing a computational analysis to identify technical variables that influence contamination patterns can help to identify species or samples to remove. However, batch-correction software should be used with caution, as these methods can introduce unconfirmed microbial signatures[171].
5. Prior knowledge and biological plausibility: certain taxa have been identified as common reagent or laboratory contaminants, such as *Ralstonia* and *Pseudomonas*[174,178,205]. Species previously reported as contaminants in low-biomass studies must be considered with excessive caution.

often processed alongside tumour samples for germline genotyping, can be used effectively to control contamination in the absence of negative controls[45]. Alternatively, biological contaminants can be removed by identifying species correlated with sequencing batches[173].

In addition to biological false positives, metagenomic analyses of tumour sequencing data must contend with technical artefacts resulting from false-positive alignments. While biological contamination can be controlled with careful handling of biological specimens, sterile technique is also required to mitigate technical false positives and avoid exposing patient genomic information[179]. Technical false-positive signals occur when human DNA or RNA sequences are erroneously aligned to microbial reference genomes that are themselves contaminated with human sequences[176,177], and aligning human sequences to these contaminated genomes can create a 'mirage', falsely suggesting the presence of a microbe. A common strategy to address this issue is computational subtraction of host sequences by alignment to the human reference genome[180]. Even so, due to the incompleteness of the human reference and genomic diversity among individuals, such approaches may not completely eliminate the issue and could even risk exposing patient-identifying genomic information[179]. Use of the updated human reference genome, T2T-CHM13, alongside rigorous screening of microbial reference genomes for human contamination[176,177] will help to mitigate these persistent challenges.

Both biological and technical false positives can be mitigated to enable the use of tumour sequencing data for microbiome analysis. Computational tools such as PathSeq[180], SEPATH[181] and PRISM[182] have been developed specifically for the purpose of identifying microorganisms in tissue sequencing data. Each implements steps to remove human sequences and reduce false positives. Other computational tools have been developed for identifying biological contamination in microbiome studies[173,175], including techniques that are tailored to large tumour sequencing datasets such as TCGA[45]. We expect that with tools like these, and the guidelines we have outlined, false-positive microbial signals can be reduced or eliminated in future analyses of tumour sequencing datasets. Many of these strategies are already being implemented in such analyses[46].

Genome sequencing has been—and will continue to be—an essential tool for microbial detection and classification. However, like all techniques, it has limitations. In the long run, studies relying on tumour sequencing data will need to have their results validated using classical techniques, including cell culture, microscopy and animal models. The application of these standardized approaches will be essential for conclusively defining the scope of microbial colonization across body sites, the pathways by which these microorganisms colonize tumour tissues or surrounding areas, and their influence on the tumour microenvironment.

Outlook

According to current global estimates, one in eight cancers diagnosed each year has an infectious origin[51]. The bulk of this risk is attributed to individual microorganisms, but there is now growing evidence that different cancer types harbour microbes, including bacteria, fungi and/or viruses. For certain cancers, the role of tumour-associated microorganisms is well supported by multiple lines of evidence, including genomic studies, culture-based methods, microscopy, epidemiological research, and through the establishment of cause–effect relationships in experimental models. Such links have been established for several viral pathogens, parasites and some bacteria, which have been shown to promote cancer both locally and systemically, through direct and indirect mechanisms.

Future enquiries into the cancer microbiome should be approached with measured reservation, and findings must undergo careful and rigorous validation before being translated into clinical practice. At the exploratory stage, cancer microbiome researchers must develop and adhere to experimental and computational guidelines and standards to ensure reproducibility.

For most cancer types, our understanding of the multi-kingdom cancer microbiome and its role in onset, progression and therapeutic response is still in its infancy. Mechanistically, the extent to which failures of key checkpoints, such as dysbiosis or an insufficient immune response, influence microbial dissemination and colonization has yet to be fully deciphered, particularly for traditionally sterile tumour sites. Understanding these colonization pathways and their implications for cancer development, progression and metastasis will be essential, not only for therapeutic targeting, but also for informing early detection and preventative strategies to reduce cancer risk.

Empirically, vaccination programmes and other public health screening practices deployed to combat oncomicrobes such as HPV, *H. pylori* and hepatitis B and C have proven highly successful in reducing global cancer incidence. Expanding efforts to target tumour-associated microorganisms has enormous potential to prevent and treat cancer development in at-risk populations. Oncomicrobes influence tumorigenesis directly and indirectly through a multiplicity of pathways, including chronic inflammation of precancerous tissue, expression of mutagenic toxins, suppression of the antitumour immune response, and promoting a metastatic phenotype, and each of these tumorigenic pathways represents a potential target for therapeutic intervention.

Conversely, there is compelling evidence that the delivery of beneficial microbial components to patients with cancer, either locally or systemically, can provide therapeutic benefits to patients. These strategies have shown particular promise in the context of immune checkpoint blockade[127,132,139,140,166,183–186]. As immunotherapies become an option for a growing number of patients, it will be essential to decipher the intricate crosstalk between the multi-kingdom cancer microbiome and tissue-resident immune cells before these findings can make their way to the clinic.

Key to achieving these endeavours will be leveraging new technologies such as spatial transcriptomics and spatial proteomics, maturing technologies like single-cell sequencing, and well-established methods for characterizing microbial communities, including cell culture, systemic mass spectrometry, and high-throughput DNA and RNA sequencing. Collectively, these tools will be essential for deciphering biogeographical colonization and dissemination patterns, as well as the local spatial organization of tumour-associated microbes and tissue-associated immune populations within the tumour microenvironment. These insights will not only deepen our understanding of cancer biology but also drive the development of novel diagnostic, therapeutic and public health strategies aimed at mitigating microbial contributions to cancer risk and progression.

References

1. Rous, P. A sarcoma of the fowl transmissible by an agent separable from the tumor cells. *J. Exp. Med.* **13**, 397–411 (1911).
2. Avery, O. T., MacLeod, C. M. & McCarty, M. Studies on the chemical nature of the substance inducing transformation of pneumococcal types: induction of transformation by a desoxyribonucleic acid fraction isolated from Pneumococcus Type III. *J. Exp. Med.* **79**, 137–158 (1944).
3. Stehelin, D., Varmus, H. E., Bishop, J. M. & Vogt, P. K. DNA related to the transforming gene(s) of avian sarcoma viruses is present in normal avian DNA. *Nature* **260**, 170–173 (1976).
4. Shih, C. & Weinberg, R. A. Isolation of a transforming sequence from a human bladder carcinoma cell line. *Cell* **29**, 161–169 (1982).
5. Reddy, E. P., Reynolds, R. K., Santos, E. & Barbacid, M. A point mutation is responsible for the acquisition of transforming properties by the T24 human bladder carcinoma oncogene. *Nature* **300**, 149–152 (1982).
6. Yoshida, M., Miyoshi, I. & Hinuma, Y. Isolation and characterization of retrovirus from cell lines of human adult T-cell leukemia and its implication in the disease. *Proc. Natl Acad. Sci. USA* **79**, 2031–2035 (1982).
7. Kalyanaraman, V. S. et al. Natural antibodies to the structural core protein (p24) of the human T-cell leukemia (lymphoma) retrovirus found in sera of leukemia patients in Japan. *Proc. Natl Acad. Sci. USA* **79**, 1653–1657 (1982).
8. Bell, T. M., Massie, A., Ross, M. G. R. & Williams, M. C. Isolation of a reovirus from a case of Burkitt's lymphoma. *Br. Med J.* **1**, 1212–1213 (1964).
9. Dürst, M., Gissmann, L., Ikenberg, H. & zur Hausen, H. A papillomavirus DNA from a cervical carcinoma and its prevalence in cancer biopsy samples from different geographic regions. *Proc. Natl Acad. Sci. USA* **80**, 3812–3815 (1983).
10. Warren, J. R. & Marshall, B. Unidentified curved bacilli on gastric epithelium in active chronic gastritis. *Lancet* **321**, 1273–1275 (1983).
11. Plummer, M., Franceschi, S., Vignat, J., Forman, D. & de Martel, C. Global burden of gastric cancer attributable to *Helicobacter pylori*. *Int. J. Cancer* **136**, 487–490 (2015).
12. Li, W.-Q. et al. Effects of *Helicobacter pylori* treatment and vitamin and garlic supplementation on gastric cancer incidence and mortality: follow-up of a randomized intervention trial. *Brit. Med. J.* **366**, l5016 (2019).
13. Ford, A. C., Yuan, Y., Forman, D., Hunt, R. & Moayyedi, P. *Helicobacter pylori* eradication for the prevention of gastric neoplasia. *Cochrane Database Syst. Rev.* **7**, CD005583 (2020).
14. Chiang, T.-H. et al. Mass eradication of *Helicobacter pylori* to reduce gastric cancer incidence and mortality: a long-term cohort study on Matsu Islands. *Gut* **70**, 243–250 (2021).
15. Sears, C. L. Enterotoxigenic *Bacteroides fragilis*: a rogue among symbiotes. *Clin. Microbiol. Rev.* **22**, 349–369 (2009).
16. El Tekle, G. & Garrett, W. S. Bacteria in cancer initiation, promotion and progression. *Nat. Rev. Cancer* **23**, 600–618 (2023).
17. Brennan, C. A. & Garrett, W. S. *Fusobacterium nucleatum*—symbiont, opportunist and oncobacterium. *Nat. Rev. Microbiol.* **17**, 156–166 (2019).

18. Sepich-Poore, G. D. et al. The microbiome and human cancer. *Science* **371**, eabc4552 (2021).
19. Iliev, I. D. et al. Focus on fungi. *Cell* **187**, 5121–5127 (2024).
20. Galloway-Peña, J., Iliev, I.ÿD. & McAllister, F. Fungi in cancer. *Nat. Rev. Cancer* **24**, 295–298 (2024).
21. Dohlman, A. B. et al. A pan-cancer mycobiome analysis reveals fungal involvement in gastrointestinal and lung tumors. *Cell* **185**, 3807–3822.e12 (2022).
22. Narunsky-Haziza, L. et al. Pan-cancer analyses reveal cancer-type-specific fungal ecologies and bacteriome interactions. *Cell* **185**, 3789–3806.e17 (2022).
23. Xavier, J. B. et al. The cancer microbiome: distinguishing direct and indirect effects requires a systemic view. *Trends Cancer* **6**, 192–204 (2020).
24. Turnbaugh, P. J. et al. The human microbiome project. *Nature* **449**, 804–810 (2007).
25. Pleguezuelos-Manzano, C. et al. Mutational signature in colorectal cancer caused by genotoxic *pks*$^+$ *E. coli*. *Nature* **580**, 269–273 (2020).
26. Dejea, C. M. et al. Patients with familial adenomatous polyposis harbor colonic biofilms containing tumorigenic bacteria. *Science* **359**, 592–597 (2018).
27. Cao, Y. et al. Commensal microbiota from patients with inflammatory bowel disease produce genotoxic metabolites. *Science* **378**, eabm3233 (2022).
28. Alam, A. et al. Fungal mycobiome drives IL-33 secretion and type 2 immunity in pancreatic cancer. *Cancer Cell* **40**, 153–167.e11 (2022).
29. Arthur, J. C. et al. Intestinal inflammation targets cancer-inducing activity of the microbiota. *Science* **338**, 120–123 (2012).
30. Cullin, N., Azevedo Antunes, C., Straussman, R., Stein-Thoeringer, C. K. & Elinav, E. Microbiome and cancer. *Cancer Cell* **39**, 1317–1341 (2021).
31. Scott, A. J. et al. International Cancer Microbiome Consortium consensus statement on the role of the human microbiome in carcinogenesis. *Gut* **68**, 1624–1632 (2019).
32. Kostic, A. D. et al. *Fusobacterium nucleatum* potentiates intestinal tumorigenesis and modulates the tumor immune microenvironment. *Cell Host Microbe* **14**, 207–215 (2013).
33. Di Domenico, E. G., Cavallo, I., Pontone, M., Toma, L. & Ensoli, F. Biofilm producing *Salmonella* Typhi: chronic colonization and development of gallbladder cancer. *Int. J. Mol. Sci.* **18**, 1887 (2017).
34. Prorok-Hamon, M. et al. Colonic mucosa-associated diffusely adherent *afaC*$^+$ *Escherichia coli* expressing *lpfA* and *pks* are increased in inflammatory bowel disease and colon cancer. *Gut* **63**, 761–770 (2014).
35. IARC Working Group on the Evaluation of Carcinogenic Risks to Humans. Biological agents. *IARC Monogr. Eval. Carcinog. Risks Hum.* **100**, 1–441 (2012).
36. Matsuoka, M. & Jeang, K.-T. Human T-cell leukaemia virus type 1 (HTLV-1) infectivity and cellular transformation. *Nat. Rev. Cancer* **7**, 270–280 (2007).
37. Boshoff, C. & Weiss, R. AIDS-related malignancies. *Nat. Rev. Cancer* **2**, 373–382 (2002).
38. Alonso-Roman, R. et al. *Lactobacillus rhamnosus* colonisation antagonizes *Candida albicans* by forcing metabolic adaptations that compromise pathogenicity. *Nat. Commun.* **13**, 3192 (2022).
39. Vazquez-Munoz, R., Thompson, A., Sobue, T. & Dongari-Bagtzoglou, A. A prebiotic diet modulates the oral microbiome composition and results in the attenuation of oropharyngeal candidiasis in mice. *Microbiol. Spectr.* **11**, e01734-23 (2023).
40. Wu, T. et al. Cellular components mediating coadherence of *Candida albicans* and *Fusobacterium nucleatum*. *J. Dent. Res.* **94**, 1432–1438 (2015).
41. Fu, K. et al. *Streptococcus anginosus* promotes gastric inflammation, atrophy and tumorigenesis in mice. *Cell* **187**, 882–896.e17 (2024).
42. Coker, O. O. et al. Mucosal microbiome dysbiosis in gastric carcinogenesis. *Gut* **67**, 1024–1032 (2018).
43. Warren, R. L. et al. Co-occurrence of anaerobic bacteria in colorectal carcinomas. *Microbiome* **1**, 16 (2013).
44. Robinson, A. V. & Allen-Vercoe, E. Strain specificity in fusobacterial co-aggregation with colorectal cancer-relevant species. *Anaerobe* **82**, 102758 (2023).
45. Dohlman, A. B. et al. The cancer microbiome atlas: a pan-cancer comparative analysis to distinguish tissue-resident microbiota from contaminants. *Cell Host Microbe* **29**, 281–298.e5 (2021).
46. Battaglia, T. W. et al. A pan-cancer analysis of the microbiome in metastatic cancer. *Cell* **187**, 2324–2335.e19 (2024).
47. Fried, B., Reddy, A. & Mayer, D. Helminths in human carcinogenesis. *Cancer Lett.* **305**, 239–249 (2011).
48. Schwartz, D. A. Helminths in the induction of cancer: *Opisthorchis viverrini*, *Clonorchis sinensis* and cholangiocarcinoma. *Trop. Geogr. Med.* **32**, 95–100 (1980).
49. Moore, P. S. & Chang, Y. Why do viruses cause cancer? Highlights of the first century of human tumour virology. *Nat. Rev. Cancer* **10**, 878–889 (2010).
50. Yu, D. & Liu, Z. The research progress in the interaction between *Candida albicans* and cancers. *Front. Microbiol.* **13**, 988734 (2022).
51. de Martel, C. et al. Global burden of cancers attributable to infections in 2008: a review and synthetic analysis. *Lancet Oncol.* **13**, 607–615 (2012).
52. Viens, L. J. Human papillomavirus-associated cancers—United States, 2008–2012. *MMWR Morb. Mortal. Wkly Rep.* **65**, 661–666 (2016).
53. Khan, G. & Hashim, M. J. Global burden of deaths from Epstein-Barr virus attributable malignancies 1990-2010. *Infect. Agent Cancer* **9**, 38 (2014).
54. Schorr, L., Mathies, M., Elinav, E. & Puschhof, J. Intracellular bacteria in cancer—prospects and debates. *npj Biofilms Microbiomes* **9**, 76 (2023).
55. Tjalsma, H., Boleij, A., Marchesi, J. R. & Dutilh, B. E. A bacterial driver–passenger model for colorectal cancer: beyond the usual suspects. *Nat. Rev. Microbiol.* **10**, 575–582 (2012).
56. Zanier, K. et al. Structural basis for hijacking of cellular LxxLL motifs by papillomavirus E6 oncoproteins. *Science* **339**, 694–698 (2013).
57. Scheffner, M., Huibregtse, J. M., Vierstra, R. D. & Howley, P. M. The HPV-16 E6 and E6-AP complex functions as a ubiquitin-protein ligase in the ubiquitination of p53. *Cell* **75**, 495–505 (1993).
58. Spurgeon, M. E. et al. Merkel cell polyomavirus large T antigen binding to pRb promotes skin hyperplasia and tumor development. *PLoS Pathog.* **18**, e1010551 (2022).
59. Global Burden of Disease Liver Cancer Collaboration. The burden of primary liver cancer and underlying etiologies from 1990 to 2015 at the global, regional and national level: results from the Global Burden of Disease Study 2015. *JAMA Oncol.* **3**, 1683–1691 (2017).
60. Dodson, R. M. et al. Intrahepatic cholangiocarcinoma: management options and emerging therapies. *J. Am. Coll. Surg.* **217**, 736–750.e4 (2013).
61. Pearce, E. J. & MacDonald, A. S. The immunobiology of schistosomiasis. *Nat. Rev. Immunol.* **2**, 499–511 (2002).
62. Mostafa, M. H., Sheweita, S. A. & O'Connor, P. J. Relationship between schistosomiasis and bladder cancer. *Clin. Microbiol. Rev.* **12**, 97–111 (1999).
63. Halawani, A. & Tamami, M. Preliminary report on the cytological diagnosis and incidence of the Bilharzia-cancer of the bladder in Egypt. *J. Egypt Med. Assoc.* **38**, 455–465 (1955).

64. Galeano Niño, J. L. et al. Effect of the intratumoral microbiota on spatial and cellular heterogeneity in cancer. *Nature* **611**, 810–817 (2022).
65. Nomburg, J. et al. An international report on bacterial communities in esophageal squamous cell carcinoma. *Int. J. Cancer* **151**, 1947–1959 (2022).
66. Parkin, D. M. The global health burden of infection-associated cancers in the year 2002. *Int. J. Cancer* **118**, 3030–3044 (2006).
67. Nejman, D. et al. The human tumor microbiome is composed of tumor type-specific intracellular bacteria. *Science* **368**, 973–980 (2020).
68. Riquelme, E. et al. Tumor microbiome diversity and composition influence pancreatic cancer outcomes. *Cell* **178**, 795–806.e12 (2019).
69. Cover, T. L. & Blaser, M. J. *Helicobacter pylori* in health and disease. *Gastroenterology* **136**, 1863–1873 (2009).
70. Yoshiyama, H. & Nakazawa, T. Unique mechanism of *Helicobacter pylori* for colonizing the gastric mucus. *Microbes Infect.* **2**, 55–60 (2000).
71. Hatakeyama, M. Structure and function of *Helicobacter pylori* CagA, the first-identified bacterial protein involved in human cancer. *Proc. Jpn Acad. Ser. B Phys. Biol. Sci.* **93**, 196–219 (2017).
72. Takahashi-Kanemitsu, A., Knight, C. T. & Hatakeyama, M. Molecular anatomy and pathogenic actions of *Helicobacter pylori* CagA that underpin gastric carcinogenesis. *Cell Mol. Immunol.* **17**, 50–63 (2020).
73. Jans, M. et al. Colibactin-driven colon cancer requires adhesin-mediated epithelial binding. *Nature* **635**, 472–480 (2024).
74. Wu, S. et al. A human colonic commensal promotes colon tumorigenesis via activation of T helper type 17 T cell responses. *Nat. Med.* **15**, 1016–1022 (2009).
75. Cuevas-Ramos, G. et al. *Escherichia coli* induces DNA damage in vivo and triggers genomic instability in mammalian cells. *Proc. Natl Acad. Sci. USA* **107**, 11537–11542 (2010).
76. Drewes, J. L. et al. Human colon cancer-derived *Clostridioides difficile* strains drive colonic tumorigenesis in mice. *Cancer Discov.* **12**, 1873–1885 (2022).
77. He, Z. et al. *Campylobacter jejuni* promotes colorectal tumorigenesis through the action of cytolethal distending toxin. *Gut* **68**, 289–300 (2019).
78. Long, X. et al. *Peptostreptococcus anaerobius* promotes colorectal carcinogenesis and modulates tumour immunity. *Nat. Microbiol.* **4**, 2319–2330 (2019).
79. Tsoi, H. et al. *Peptostreptococcus anaerobius* induces intracellular cholesterol biosynthesis in colon cells to induce proliferation and causes dysplasia in mice. *Gastroenterology* **152**, 1419–1433.e5 (2017).
80. Rubinstein, M. R. et al. *Fusobacterium nucleatum* promotes colorectal carcinogenesis by modulating E-cadherin/β-catenin signaling via its FadA adhesin. *Cell Host Microbe* **14**, 195–206 (2013).
81. Gur, C. et al. Binding of the Fap2 protein of *Fusobacterium nucleatum* to human inhibitory receptor TIGIT protects tumors from immune cell attack. *Immunity* **42**, 344–355 (2015).
82. Bullman, S. et al. Analysis of *Fusobacterium* persistence and antibiotic response in colorectal cancer. *Science* **358**, 1443–1448 (2017).
83. Zepeda-Rivera, M. et al. A distinct *Fusobacterium nucleatum* clade dominates the colorectal cancer niche. *Nature* https://doi.org/10.1038/s41586-024-07182-w (2024).
84. Gur, C. et al. *Fusobacterium nucleatum* supresses anti-tumor immunity by activating CEACAM1. *Oncoimmunology* **8**, e1581531 (2019).
85. Randi, G., Franceschi, S. & La Vecchia, C. Gallbladder cancer worldwide: geographical distribution and risk factors. *Int. J. Cancer* **118**, 1591–1602 (2006).
86. Geller, L. T. et al. Potential role of intratumor bacteria in mediating tumor resistance to the chemotherapeutic drug gemcitabine. *Science* **357**, 1156–1160 (2017).
87. Eckhoff, A. M. et al. Comprehensive assessment of the intrinsic pancreatic microbiome. *Ann. Surg.* https://doi.org/10.1097/sla.0000000000006299 (2024).
88. Littman, A. J., Jackson, L. A. & Vaughan, T. L. *Chlamydia pneumoniae* and lung cancer: epidemiologic evidence. *Cancer Epidemiol. Biomarkers Prev.* **14**, 773–778 (2005).
89. Jin, C. et al. Commensal microbiota promote lung cancer development via γδ T cells. *Cell* **176**, 998–1013.e16 (2019).
90. Stone, J. K. et al. *Acidovorax temperans* skews neutrophil maturation and polarizes Th17 cells to promote lung adenocarcinoma development. *Oncogenesis* **13**, 13 (2024).
91. Tsay, J.-C. J. et al. Lower airway dysbiosis affects lung cancer progression. *Cancer Discov.* **11**, 293–307 (2021).
92. Zhu, G. et al. Intratumour microbiome associated with the infiltration of cytotoxic CD8. T cells and patient survival in cutaneous melanoma. *Eur J. Cancer* **151**, 25–34 (2021).
93. Fu, A. et al. Tumor-resident intracellular microbiota promotes metastatic colonization in breast cancer. *Cell* **185**, 1356–1372.e26 (2022).
94. Wang, M. et al. Killing tumor-associated bacteria with a liposomal antibiotic generates neoantigens that induce anti-tumor immune responses. *Nat. Biotechnol.* https://doi.org/10.1038/s41587-023-01957-8 (2023).
95. Wang, X. et al. *Fusobacterium nucleatum* facilitates anti-PD-1 therapy in microsatellite stable colorectal cancer. *Cancer Cell* **42**, 1729–1746 (2024).
96. Gao, Y. et al. *Fusobacterium nucleatum* enhances the efficacy of PD-L1 blockade in colorectal cancer. *Signal Transduct. Target Ther.* **6**, 398 (2021).
97. Scott, B. B. & Jenkins, D. Gastro-oesophageal candidiasis. *Gut* **23**, 137–139 (1982).
98. Chang, F., Syrjänen, S., Wang, L. & Syrjänen, K. Infectious agents in the etiology of esophageal cancer. *Gastroenterology* **103**, 1336–1348 (1992).
99. Yang, C. S. Research on esophageal cancer in China: a review. *Cancer Res.* **40**, 2633–2644 (1980).
100. Zhu, F. et al. Autoreactive T cells and chronic fungal infection drive esophageal carcinogenesis. *Cell Host Microbe* **21**, 478–493 e7 (2017).
101. Wang, T. et al. The adaptor protein CARD9 protects against colon cancer by restricting mycobiota-mediated expansion of myeloid-derived suppressor cells. *Immunity* **49**, 504–514.e4 (2018).
102. Nieminen, M. T., Uittamo, J., Salaspuro, M. & Rautemaa, R. Acetaldehyde production from ethanol and glucose by non-*Candida albicans* yeasts in vitro. *Oral Oncol.* **45**, e245–e248 (2009).
103. Moyes, D. L. et al. Candidalysin is a fungal peptide toxin critical for mucosal infection. *Nature* **532**, 64–68 (2016).
104. Li, X. V. et al. Immune regulation by fungal strain diversity in inflammatory bowel disease. *Nature* **603**, 672–678 (2022).
105. Malik, A. et al. SYK-CARD9 signaling axis promotes gut fungi-mediated inflammasome activation to restrict colitis and colon cancer. *Immunity* **49**, 515–530.e5 (2018).
106. Daley, D. et al. Dectin 1 activation on macrophages by galectin 9 promotes pancreatic carcinoma and peritumoral immune tolerance. *Nat. Med.* **23**, 556–567 (2017).
107. Fletcher, A. A., Kelly, M. S., Eckhoff, A. M. & Allen, P. J. Revisiting the intrinsic mycobiome in pancreatic cancer. *Nature* **620**, E1–E6 (2023).
108. Xu, F., Saxena, D., Pushalkar, S. & Miller, G. Reply to: Revisiting the intrinsic mycobiome in pancreatic cancer. *Nature* **620**, E7–E9 (2023).

109. Fu, A., Yao, B., Dong, T. & Cai, S. Emerging roles of intratumor microbiota in cancer metastasis. *Trends Cell Biol.* **33**, 583–593 (2023).
110. Chaffer, C. L. & Weinberg, R. A. A perspective on cancer cell metastasis. *Science* **331**, 1559–1564 (2011).
111. Welch, D. R. & Hurst, D. R. Defining the hallmarks of metastasis. *Cancer Res.* **79**, 3011–3027 (2019).
112. Fares, J., Fares, M. Y., Khachfe, H. H., Salhab, H. A. & Fares, Y. Molecular principles of metastasis: a hallmark of cancer revisited. *Signal Transduct. Target Ther.* **5**, 28 (2020).
113. Wu, Y. et al. Fn-Dps, a novel virulence factor of *Fusobacterium nucleatum*, disrupts erythrocytes and promotes metastasis in colorectal cancer. *PLoS Pathog.* **19**, e1011096 (2023).
114. Li, W. T. et al. The bladder microbiome is associated with epithelial-mesenchymal transition in muscle invasive urothelial bladder carcinoma. *Cancers* **13**, 3649 (2021).
115. Lee, J. et al. Human primary epithelial cells acquire an epithelial-mesenchymal-transition phenotype during long-term infection by the oral opportunistic pathogen, *Porphyromonas gingivalis*. *Front. Cell. Infect. Microbiol.* **7**, 493 (2017).
116. Bessède, E. et al. *Helicobacter pylori* generates cells with cancer stem cell properties via epithelial-mesenchymal transition-like changes. *Oncogene* **33**, 4123–4131 (2014).
117. Parida, S. et al. A procarcinogenic colon microbe promotes breast tumorigenesis and metastatic progression and concomitantly activates notch and β-catenin axes. *Cancer Discov.* **11**, 1138–1157 (2021).
118. Zhang, Y. et al. *Fusobacterium nucleatum* promotes colorectal cancer cells adhesion to endothelial cells and facilitates extravasation and metastasis by inducing ALPK1/NF-κB/ICAM1 axis. *Gut Microbes* **14**, 2038852 (2022).
119. Bertocchi, A. et al. Gut vascular barrier impairment leads to intestinal bacteria dissemination and colorectal cancer metastasis to liver. *Cancer Cell* **39**, 708–724.e11 (2021).
120. He, Z. et al. *Campylobacter jejuni*-derived cytolethal distending toxin promotes colorectal cancer metastasis. *Cell Host Microbe* **32**, 2080–2091.e6 (2024).
121. Parhi, L. et al. Breast cancer colonization by *Fusobacterium nucleatum* accelerates tumor growth and metastatic progression. *Nat. Commun.* **11**, 3259 (2020).
122. Shang, A. et al. Exosomal circPACRGL promotes progression of colorectal cancer via the miR-142-3p/miR-506-3p-TGF-β1 axis. *Mol. Cancer* **19**, 117 (2020).
123. Channon, L. M. et al. Small extracellular vesicles (exosomes) and their cargo in pancreatic cancer: key roles in the hallmarks of cancer. *Biochim. Biophys. Acta Rev. Cancer* **1877**, 188728 (2022).
124. Li, G. et al. *Fusobacterium nucleatum*-derived small extracellular vesicles facilitate tumor growth and metastasis via TLR4 in breast cancer. *BMC Cancer* **23**, 473 (2023).
125. Gopalakrishnan, V., Helmink, B. A., Spencer, C. N., Reuben, A. & Wargo, J. A. The influence of the gut microbiome on cancer, immunity and cancer immunotherapy. *Cancer Cell* **33**, 570–580 (2018).
126. Derosa, L. et al. Custom scoring based on ecological topology of gut microbiota associated with cancer immunotherapy outcome. *Cell* **187**, 3373–3389.e16 (2024).
127. Derosa, L. et al. Intestinal *Akkermansia muciniphila* predicts clinical response to PD-1 blockade in patients with advanced non-small-cell lung cancer. *Nat. Med.* **28**, 315–324 (2022).
128. Bae, M. et al. *Akkermansia muciniphila* phospholipid induces homeostatic immune responses. *Nature* **608**, 168–173 (2022).
129. Jin, Y. et al. The diversity of gut microbiome is associated with favorable responses to anti-programmed death 1 immunotherapy in Chinese patients with NSCLC. *J. Thorac. Oncol.* **14**, 1378–1389 (2019).
130. Gopalakrishnan, V. et al. Gut microbiome modulates response to anti-PD-1 immunotherapy in melanoma patients. *Science* **359**, 97–103 (2018).
131. Giampazolias, E. et al. Vitamin D regulates microbiome-dependent cancer immunity. *Science* **384**, 428–437 (2024).
132. Vétizou, M. et al. Anticancer immunotherapy by CTLA-4 blockade relies on the gut microbiota. *Science* **350**, 1079–1084 (2015).
133. Derosa, L. et al. Gut bacteria composition drives primary resistance to cancer immunotherapy in renal cell carcinoma patients. *Eur. Urol.* **78**, 195–206 (2020).
134. Fidelle, M. et al. A microbiota-modulated checkpoint directs immunosuppressive intestinal T cells into cancers. *Science* **380**, eabo2296 (2023).
135. Xu, X. et al. Gut microbiome influences the efficacy of PD-1 antibody immunotherapy on MSS-type colorectal cancer via metabolic pathway. *Front. Microbiol.* **11**, 814 (2020).
136. Baruch, E. N. et al. Fecal microbiota transplant promotes response in immunotherapy-refractory melanoma patients. *Science* **371**, 602–609 (2021).
137. Spencer, C. N. et al. Dietary fiber and probiotics influence the gut microbiome and melanoma immunotherapy response. *Science* **374**, 1632–1640 (2021).
138. Lionakis, M. S., Drummond, R. A. & Hohl, T. M. Immune responses to human fungal pathogens and therapeutic prospects. *Nat. Rev. Immunol.* **23**, 433–452 (2023).
139. Sivan, A. et al. Commensal *Bifidobacterium* promotes antitumor immunity and facilitates anti-PD-L1 efficacy. *Science* **350**, 1084–1089 (2015).
140. Davar, D. et al. Fecal microbiota transplant overcomes resistance to anti-PD-1 therapy in melanoma patients. *Science* **371**, 595–602 (2021).
141. Zagato, E. et al. Endogenous murine microbiota member *Faecalibaculum rodentium* and its human homologue protect from intestinal tumour growth. *Nat. Microbiol.* **5**, 511–524 (2020).
142. Starnes, C. O. Coley's toxins in perspective. *Nature* **357**, 11–12 (1992).
143. Hoption Cann, S. A., van Netten, J. P. & van Netten, C. Dr William Coley and tumour regression: a place in history or in the future. *Postgrad. Med. J.* **79**, 672–680 (2003).
144. Morales, A., Eidinger, D. & Bruce, A. W. Intracavitary *Bacillus* Calmette-Guerin in the treatment of superficial bladder tumors. *J. Urol.* **116**, 180–183 (1976).
145. Redelman-Sidi, G., Glickman, M. S. & Bochner, B. H. The mechanism of action of BCG therapy for bladder cancer—a current perspective. *Nat. Rev. Urol.* **11**, 153–162 (2014).
146. Conry, R. M. et al. Talimogene laherparepvec: first in class oncolytic virotherapy. *Hum. Vaccin. Immunother* **14**, 839–846 (2018).
147. Boorjian, S. A. et al. Intravesical nadofaragene firadenovec gene therapy for BCG-unresponsive non-muscle-invasive bladder cancer: a single-arm, open-label, repeat-dose clinical trial. *Lancet Oncol.* **22**, 107–117 (2021).
148. Forbes, N. S. Engineering the perfect (bacterial) cancer therapy. *Nat. Rev. Cancer* **10**, 785–794 (2010).
149. Zhou, S. et al. Tumour-targeting bacteria engineered to fight cancer. *Nat. Rev. Cancer* **18**, 727–743 (2018).
150. Nguyen, D.-H., Chong, A., Hong, Y. & Min, J.-J. Bioengineering of bacteria for cancer immunotherapy. *Nat. Commun.* **14**, 3553 (2023).
151. Kasinskas, R. W. & Forbes, N. S. *Salmonella typhimurium* lacking ribose chemoreceptors localize in tumor quiescence and induce apoptosis. *Cancer Res.* **67**, 3201–3209 (2007).
152. Pawelek, J. M., Low, K. B. & Bermudes, D. Tumor-targeted *Salmonella* as a novel anticancer vector. *Cancer Res.* **57**, 4537–4544 (1997).

153. Gurbatri, C.ÿR., Arpaia, N. & Danino, T. Engineering bacteria as interactive cancer therapies. *Science* **378**, 858–864 (2022).
154. Zheng, J. H. et al. Two-step enhanced cancer immunotherapy with engineered *Salmonella typhimurium* secreting heterologous flagellin. *Sci. Transl. Med.* **9**, eaak9537 (2017).
155. Chabloz, A. et al. *Salmonella*-based platform for efficient delivery of functional binding proteins to the cytosol. *Commun. Biol.* **3**, 342 (2020).
156. Camacho, E. M., Mesa-Pereira, B., Medina, C., Flores, A. & Santero, E. Engineering *Salmonella* as intracellular factory for effective killing of tumour cells. *Sci. Rep.* **6**, 30591 (2016).
157. Raman, V. et al. Intracellular delivery of protein drugs with an autonomously lysing bacterial system reduces tumor growth and metastases. *Nat. Commun.* **12**, 6116 (2021).
158. Din, M. O. et al. Synchronized cycles of bacterial lysis for in vivo delivery. *Nature* **536**, 81–85 (2016).
159. Yang, S. et al. Non-pathogenic *E. coli* displaying decoy-resistant IL18 mutein boosts anti-tumor and CAR NK cell responses. *Nat. Biotechnol.* **43**, 1311–1323 (2025).
160. Canale, F. P. Metabolic modulation of tumours with engineered bacteria for immunotherapy. *Nature* **598**, 662–666 (2021).
161. Chen, Y. E. et al. Engineered skin bacteria induce antitumor T cell responses against melanoma. *Science* **380**, 203–210 (2023).
162. Kostic, A. D. et al. Genomic analysis identifies association of *Fusobacterium* with colorectal carcinoma. *Genome Res.* **22**, 292–298 (2012).
163. Castellarin, M. et al. *Fusobacterium nucleatum* infection is prevalent in human colorectal carcinoma. *Genome Res.* **22**, 299–306 (2012).
164. Roelands, J. et al. An integrated tumor, immune and microbiome atlas of colon cancer. *Nat. Med* **29**, 1273–1286 (2023).
165. Abed, J. et al. Colon cancer-associated *Fusobacterium nucleatum* may originate from the oral cavity and reach colon tumors via the circulatory system. *Front. Cell. Infect. Microbiol.* **10**, 400 (2020).
166. Choi, Y. et al. Immune checkpoint blockade induces gut microbiota translocation that augments extraintestinal antitumor immunity. *Sci. Immunol.* **8**, eabo2003 (2023).
167. Tan, C. C. S. et al. No evidence for a common blood microbiome based on a population study of 9,770 healthy humans. *Nat. Microbiol.* **8**, 973–985 (2023).
168. Ge, Y., Lu, J., Puiu, D., Revsine, M. & Salzberg, S. L. Comprehensive analysis of microbial content in whole-genome sequencing samples from The Cancer Genome Atlas project. Preprint at *bioRxiv* https://doi.org/10.1101/2024.05.24.595788 (2024).
169. Gihawi, A., Cooper, C. S. & Brewer, D. S. Caution regarding the specificities of pan-cancer microbial structure. *Microb. Genom.* **9**, mgen001088 (2023).
170. Sepich-Poore, G. D. et al. Reply to: caution regarding the specificities of pan-cancer microbial structure. Preprint at *bioRxiv* https://doi.org/10.1101/2023.02.10.528049 (2023).
171. Gihawi, A. et al. Major data analysis errors invalidate cancer microbiome findings. *mBio* **14**, e01607-23 (2023).
172. Sepich-Poore, G. D. et al. Robustness of cancer microbiome signals over a broad range of methodological variation. *Oncogene* **43**, 1127–1148 (2024).
173. Austin, G. I. et al. Contamination source modeling with SCRuB improves cancer phenotype prediction from microbiome data. *Nat. Biotechnol.* **41**, 1820–1828 (2023).
174. Eisenhofer, R. et al. Contamination in low microbial biomass microbiome studies: issues and recommendations. *Trends Microbiol.* **27**, 105–117 (2019).
175. Davis, N. M., Proctor, D. M., Holmes, S. P., Relman, D. A. & Callahan, B. J. Simple statistical identification and removal of contaminant sequences in marker-gene and metagenomics data. *Microbiome* **6**, 226 (2018).
176. Chrisman, B. et al. The human 'contaminome': bacterial, viral and computational contamination in whole genome sequences from 1,000 families. *Sci. Rep.* **12**, 9863 (2022).
177. Breitwieser, F. P., Pertea, M., Zimin, A. V. & Salzberg, S. L. Human contamination in bacterial genomes has created thousands of spurious proteins. *Genome Res.* **29**, 954–960 (2019).
178. Fierer, N. et al. Guidelines for preventing and reporting contamination in low biomass microbiome studies. *Nat. Microbiol.* **10**, 1570–1580 (2025).
179. Guccione, C. et al. Incomplete human reference genomes can drive false sex biases and expose patient-identifying information in metagenomic data. *Nat. Commun.* **16**, 825 (2025).
180. Kostic, A. D. et al. PathSeq: a comprehensive computational tool for the identification or discovery of microorganisms by deep sequencing of human tissue. *Nat. Biotechnol.* **29**, 393–396 (2011).
181. Gihawi, A. et al. SEPATH: benchmarking the search for pathogens in human tissue whole genome sequence data leads to template pipelines. *Genome Biol.* **20**, 208 (2019).
182. Ghaddar, B. C., Blaser, M. J. & De, S. Revisiting the cancer microbiome using PRISM. Preprint at *bioRxiv* https://doi.org/10.1101/2025.01.21.634087 (2025).
183. Matson, V. et al. The commensal microbiome is associated with anti-PD-1 efficacy in metastatic melanoma patients. *Science* **359**, 104–108 (2018).
184. Routy, B. et al. Gut microbiome influences efficacy of PD-1-based immunotherapy against epithelial tumors. *Science* **359**, 91–97 (2018).
185. Griffin, M. E. et al. *Enterococcus* peptidoglycan remodeling promotes checkpoint inhibitor cancer immunotherapy. *Science* **373**, 1040–1046 (2021).
186. Viaud, S. et al. The intestinal microbiota modulates the anticancer immune effects of cyclophosphamide. *Science* **342**, 971–976 (2013).
187. McIlvanna, E., Linden, G. J., Craig, S. G., Lundy, F. T. & James, J. A. *Fusobacterium nucleatum* and oral cancer: a critical review. *BMC Cancer* **21**, 1212 (2021).
188. Yamamura, K. et al. Human microbiome *Fusobacterium nucleatum* in esophageal cancer tissue is associated with prognosis. *Clin. Cancer Res.* **22**, 5574–5581 (2016).
189. The Cancer Genome Atlas Research Network Comprehensive molecular characterization of gastric adenocarcinoma. *Nature* **513**, 202–209 (2014).
190. Boleij, A. et al. The *Bacteroides fragilis* toxin gene is prevalent in the colon mucosa of colorectal cancer patients. *Clin. Infect. Dis.* **60**, 208–215 (2015).
191. Wilson, M. R. et al. The human gut bacterial genotoxin colibactin alkylates DNA. *Science* **363**, eaar7785 (2019).
192. Zhao, L. et al. *Parvimonas micra* promotes colorectal tumorigenesis and is associated with prognosis of colorectal cancer patients. *Oncogene* **41**, 4200–4210 (2022).
193. Chang, Y. et al. *Parvimonas micra* activates the Ras/ERK/c-Fos pathway by upregulating miR-218-5p to promote colorectal cancer progression. *J. Exp. Clin. Cancer Res.* **42**, 13 (2023).
194. Liu, Y. et al. *Peptostreptococcus anaerobius* mediates anti-PD1 therapy resistance and exacerbates colorectal cancer via myeloid-derived suppressor cells in mice. *Nat. Microbiol.* **9**, 1467–1482 (2024).
195. Feng, H., Shuda, M., Chang, Y. & Moore, P. S. Clonal integration of a polyomavirus in human Merkel cell carcinoma. *Science* **319**, 1096–1100 (2008).
196. Rencic, A. et al. Detection of JC virus DNA sequence and expression of the viral oncoprotein, tumor antigen, in brain of immunocompetent patient with oligoastrocytoma. *Proc. Natl Acad. Sci. USA* **93**, 7352–7357 (1996).
197. Del Valle, L. et al. Detection of JC virus DNA sequences and expression of the viral regulatory protein T-antigen in tumors of the central nervous system. *Cancer Res.* **61**, 4287–4293 (2001).

198. Rollison, D. E. M. et al. Serum antibodies to JC virus, BK virus, simian virus 40, and the risk of incident adult astrocytic brain tumors. *Cancer Epidemiol. Biomark. Prev.* **12**, 460–463 (2003).
199. Liu, N.-N. et al. The intratumor mycobiome promotes lung cancer progression via myeloid-derived suppressor cells. *Cancer Cell* **41**, 1927–1944.e9 (2023).
200. Stroffolini, T. & Stroffolini, G. A historical overview on the role of Hepatitis B and C viruses as aetiological factors for hepatocellular carcinoma. *Cancers* **15**, 2388 (2023).
201. Aykut, B. et al. The fungal mycobiome promotes pancreatic oncogenesis via activation of MBL. *Nature* **574**, 264–267 (2019).
202. Nikou, S.-A. et al. The *Candida albicans* toxin candidalysin mediates distinct epithelial inflammatory responses through p38 and EGFR-ERK pathways. *Sci. Signal.* **15**, eabj6915 (2022).
203. Ho, J. et al. Candidalysin activates innate epithelial immune responses via epidermal growth factor receptor. *Nat. Commun.* **10**, 2297 (2019).
204. Breitwieser, F. P., Baker, D. N. & Salzberg, S. L. KrakenUniq: confident and fast metagenomics classification using unique k-mer counts. *Genome Biol.* **19**, 198 (2018).
205. Salter, S. J. et al. Reagent and laboratory contamination can critically impact sequence-based microbiome analyses. *BMC Biol.* **12**, 87 (2014).

Acknowledgements

We acknowledge support by The Damon Runyon Cancer Research Foundation (DRG-2504-23, to A.B.D.) and the US National Institutes of Health (R01CA286920, R01DK113136, R01DK121977, R01 AI178683 and R01AI163007), the Leona M. and Harry B. Helmsley Charitable Trust, the Irma T. Hirschl Career Scientist, the Research Corporation for Science Advancement, the Kenneth Rainin Innovator Award, the Burroughs Wellcome Fund PATH Fellow Award, the CIFAR program Fungal Kingdom: Threats and Opportunities Fellow and the Cancer Research Institute and Lloyd J. Old STAR Fellow Awards (to I.D.I.).

Competing interests

A.B.D. and I.D.I. are inventors on US patent application no. PCT/US2023/029634, submitted by Cornell University and Duke University, related to the use of fungi for classifying and treating gastrointestinal and lung tumours. L.Z. is founder of a Biotech. Cie involved in the cancer/microbiome space everImmune; is president of the everImmune scientific advisory board (SAB); has obtained research funding with Pileje, Biomérieux, everImmune and Daiichi Sankyo; is a SAB member of the DKFZ and German Cancer Aid; has received honoraria from everImmune; is an employee of Gustave Roussy and the Faculté de Medecine Paris XI; is ex editor in chief of *OncoImmunology*; is an ex member of the SAB of EpiVax, Lytix Biopharma and Tusk Therapeutics; has past research contracts with Kaleido, Elior, Carrefour, Merus, Roche, Tusk, Pileje, Incyte, BMS, GSK, Lytix Biopharma, Innate Pharma and Transgene; is an ex member of the board of directors of Transgene; is an ex member of the SAB of Hookipa; had a teaching contract with Pierre Fabre in 2023; and is ex president of the SAB of IHU Méditerranée Infections. I.D.I. has received honoraria from Moderna. X.P. declares no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Anders B. Dohlman or Iliyan D. Iliev.

Peer review information *Nature Microbiology* thanks Maria Rescigno and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© Springer Nature Limited 2025